

Assistant: Hello! I'm an AI assistant here to help you narrow down the Material requirements for Grade selection. Could you please tell me which application you're considering?

User: I need a material for a rogowski type sensor

Assistant: To help identify the best material for your Rogowski type sensor, could you please clarify:

1. What is the exact use of the polymer in your sensor? For example, is it for the flexible jacket around the coil, the sensor housing, or another component?
2. Where will this sensor be used? For instance, is it for EV charging stations, industrial switchgear, automotive electrical systems, or another application?

This will help ensure the material suggestion fits your specific requirements.

User: jacket for utility monitoring

Assistant: Thank you for clarifying. To ensure the material suggestion fits your needs, could you please provide:

1. What is the typical operating environment for your Rogowski coil jacket in utility monitoring? For example, will it be used indoors (protected from weather) or outdoors (exposed to UV, rain, temperature extremes, or chemicals)?
2. What is the minimum wall thickness of the jacket?
3. Are there any preferred base materials you have used previously or would like to use for this application?
4. Are you open to alternative material suggestions if there is no strict preference for

the base material?

These details will help identify the most suitable material options for your application.

User: UL94 V0 at 1.6mm no preference open to suggestions

Assistant: Here are the key material requirements (CTQs) assumed for your Rogowski coil jacket application:

1. Minimum wall thickness: 1.6 mm
2. Flame rating: UL94 V-0 at 1.6 mm
3. Environmental exposure: Flame retardant
4. Environmental exposure: UV stabilized or stable to weather
5. Processing requirement: Extrusion - Wire and Cable
6. Mechanical durability: Flexing/flex-fatigue resistance

Are all of these requirements correct for your application? Please confirm if these match your needs.

User: i also need it to be halogen free and Shore A hardness 80

Assistant: Thank you for specifying halogen-free and Shore A hardness 80. To finalize your requirements, could you please confirm:

1. Halogen-free: Required
2. Shore A hardness: 80
3. UL Yellow Card Property: Outdoor suitability (f1 rating)—Is this required for your application, given possible outdoor exposure?

Please confirm or clarify these requirements so I can identify the most suitable material

options for your Rogowski coil jacket.

User: it can face outdoor exposure

Assistant: Based on your requirements, here are the closest material options for your Rogowski coil jacket:

1. SANTOPRENE® 251-90 HF: Halogen-free, UL94 V-0 at 1.6 mm, wire & cable extrusion, flex-fatigue resistance. Shore A hardness is 94 (stiffer than your target of 80). Outdoor suitability (UL f1) is not explicitly listed and may require confirmation.
2. SANTOPRENE® 151-70W256: UL f1 outdoor suitability, Shore A hardness ~75 (close to 80), wire & cable extrusion, flex-fatigue resistance. Flame rating is UL94 V-1 at 1.6 mm (not V-0), and halogen-free status is not explicitly stated.
3. Hytrel® HTR8908ECO-B 452 NC010: Halogen-free, Shore A ~90, wire & cable extrusion, flex-fatigue resistance. UL94 V-0 only at ≥ 3.1 mm (not at 1.6 mm), f1 not listed.
4. Hytrel® HTR8813 BK320: Halogen-free, UL94 V-0 at 1.5 & 3.0 mm, wire & cable extrusion, flex-fatigue resistance. Hardness is higher (~Shore A 98), f1 not listed.
5. SANTOPRENE® 253-40 HF: Halogen-free, UL94 V-0 at 1.5 mm, wire & cable extrusion, flex-fatigue resistance. Hardness is much higher (Shore D 46), f1 not listed.

None of these grades meet all requirements exactly, so a trade-off may be needed (e.g., accepting higher hardness or V-1 flame rating). These options may be suitable for your application, but final material selection and product design decisions rest with you. Please ensure compliance with all applicable standards and regulations. Celanese does not manufacture or certify products for these applications. If you need further confirmation on outdoor suitability or halogen-free status, I recommend connecting with a live Celanese materials specialist.

Note: This recommendation is generated by an AI assistant and should be considered purely informational.